

Unit 2: Limit of a Function and Limit Laws — Full Solutions

Warm-up

1. $\lim_{x \rightarrow 3} (2x + 1)$

This is a straight line — no division, no square roots, nothing tricky. Just plug in $x = 3$:

$$2(3) + 1 = 7$$

Answer: 7.

2. $\lim_{x \rightarrow -2} (x^2 - 3x + 4)$

Plug in $x = -2$ directly:

$$(-2)^2 - 3(-2) + 4 = 4 + 6 + 4 = 14$$

Answer: 14.

3. $\lim_{x \rightarrow 1} (3x^2 - 2x + 5)$

Plug in $x = 1$:

$$3(1)^2 - 2(1) + 5 = 3 - 2 + 5 = 6$$

Answer: 6.

4. $\lim_{x \rightarrow 0} (x^3 + 2x^2 - 1)$

Plug in $x = 0$:

$$0 + 0 - 1 = -1$$

Answer: -1 .

5. $\lim_{x \rightarrow 4} \sqrt{x + 5}$

Plug in $x = 4$:

$$\sqrt{4 + 5} = \sqrt{9} = 3$$

Answer: 3.

6. $\lim_{x \rightarrow 2} \frac{x^2 + 1}{x + 3}$

The denominator at $x = 2$ is $2 + 3 = 5$, which is not zero, so we can just plug straight in:

$$\frac{2^2 + 1}{2 + 3} = \frac{5}{5} = 1$$

Answer: 1.

$$7. \lim_{x \rightarrow 5} \frac{x+5}{x^2-20}$$

The denominator at $x = 5$ is $25 - 20 = 5$, not zero, so direct substitution works:

$$\frac{5+5}{25-20} = \frac{10}{5} = 2$$

Answer: 2.

Standard

$$8. \lim_{x \rightarrow 2} \frac{x^2 - 7x + 10}{x - 2}$$

Plugging in $x = 2$ gives $\frac{4-14+10}{0} = \frac{0}{0}$ — a signal to factor.

Factor the top: we need two numbers that multiply to 10 and add to -7 : those are -2 and -5 .

$$x^2 - 7x + 10 = (x-2)(x-5)$$

So the fraction becomes:

$$\frac{(x-2)(x-5)}{x-2}$$

Cancel the common factor $(x-2)$ (allowed since we're looking at values near, not at, $x = 2$):

$$x-5$$

Now plug in $x = 2$: $2 - 5 = -3$.

Answer: -3 .

$$9. \lim_{x \rightarrow 3} \frac{x^2 - 9}{x - 3}$$

Plugging in gives $\frac{0}{0}$. Factor the top as a difference of squares:

$$x^2 - 9 = (x-3)(x+3)$$

Cancel $(x-3)$:

$$x+3$$

Plug in $x = 3$: $3 + 3 = 6$.

Answer: 6.

$$10. \lim_{x \rightarrow -1} \frac{x^2 + 3x + 2}{x + 1}$$

Plugging in gives $\frac{0}{0}$. Factor the top: need two numbers multiplying to 2, adding to 3: those are 1 and 2.

$$x^2 + 3x + 2 = (x + 1)(x + 2)$$

Cancel $(x + 1)$:

$$x + 2$$

Plug in $x = -1$: $-1 + 2 = 1$.

Answer: 1.

$$11. \lim_{x \rightarrow 4} \frac{x^2 - 16}{x - 4}$$

Plugging in gives $\frac{0}{0}$. Factor as difference of squares:

$$x^2 - 16 = (x - 4)(x + 4)$$

Cancel $(x - 4)$:

$$x + 4$$

Plug in $x = 4$: $4 + 4 = 8$.

Answer: 8.

$$12. \lim_{x \rightarrow 1} \frac{x^3 - 1}{x - 1}$$

Plugging in gives $\frac{0}{0}$. Use the “difference of cubes” pattern: $a^3 - b^3 = (a - b)(a^2 + ab + b^2)$, with $a = x$, $b = 1$:

$$x^3 - 1 = (x - 1)(x^2 + x + 1)$$

Cancel $(x - 1)$:

$$x^2 + x + 1$$

Plug in $x = 1$: $1 + 1 + 1 = 3$.

Answer: 3.

$$13. \lim_{x \rightarrow 2} \frac{x^2 - 5x + 6}{x^2 - 4}$$

Plugging in gives $\frac{4 - 10 + 6}{4 - 4} = \frac{0}{0}$.

Factor the top: numbers multiplying to 6, adding to -5 : -2 and -3 .

$$x^2 - 5x + 6 = (x - 2)(x - 3)$$

Factor the bottom as a difference of squares:

$$x^2 - 4 = (x - 2)(x + 2)$$

So the fraction is:

$$\frac{(x - 2)(x - 3)}{(x - 2)(x + 2)}$$

Cancel $(x - 2)$:

$$\frac{x - 3}{x + 2}$$

Plug in $x = 2$: $\frac{2 - 3}{2 + 2} = \frac{-1}{4}$.

Answer: $-\frac{1}{4}$.

14. $\lim_{x \rightarrow 3} \frac{x^2 - 2x - 3}{x^2 - 9}$

Plugging in gives $\frac{9 - 6 - 3}{0} = \frac{0}{0}$.

Factor the top: numbers multiplying to -3 , adding to -2 : -3 and 1 .

$$x^2 - 2x - 3 = (x - 3)(x + 1)$$

Factor the bottom:

$$x^2 - 9 = (x - 3)(x + 3)$$

Cancel $(x - 3)$:

$$\frac{x + 1}{x + 3}$$

Plug in $x = 3$: $\frac{3 + 1}{3 + 3} = \frac{4}{6} = \frac{2}{3}$.

Answer: $\frac{2}{3}$.

Challenge

$$15. \lim_{x \rightarrow 4} \frac{x-4}{\sqrt{x}-2}$$

Plugging in gives $\frac{0}{0}$. There's a square root in the denominator, so we rationalize by multiplying top and bottom by the conjugate $\sqrt{x}+2$:

$$\frac{x-4}{\sqrt{x}-2} \cdot \frac{\sqrt{x}+2}{\sqrt{x}+2} = \frac{(x-4)(\sqrt{x}+2)}{(\sqrt{x}-2)(\sqrt{x}+2)}$$

The denominator becomes $(\sqrt{x})^2 - 2^2 = x - 4$ (this is exactly why we picked the conjugate — it clears the square root).

$$\frac{(x-4)(\sqrt{x}+2)}{x-4}$$

Cancel $(x-4)$:

$$\sqrt{x}+2$$

Plug in $x = 4$: $\sqrt{4}+2 = 2+2 = 4$.

Answer: 4.

$$16. \lim_{x \rightarrow 2} \frac{\sqrt{x^2+12}-4}{x-2}$$

Plugging in gives $\frac{0}{0}$. Rationalize by multiplying by the conjugate $\sqrt{x^2+12}+4$:

$$\frac{\sqrt{x^2+12}-4}{x-2} \cdot \frac{\sqrt{x^2+12}+4}{\sqrt{x^2+12}+4} = \frac{(x^2+12)-16}{(x-2)(\sqrt{x^2+12}+4)}$$

Simplify the top: $x^2+12-16 = x^2-4 = (x-2)(x+2)$.

$$\frac{(x-2)(x+2)}{(x-2)(\sqrt{x^2+12}+4)}$$

Cancel $(x-2)$:

$$\frac{x+2}{\sqrt{x^2+12}+4}$$

Plug in $x = 2$: $\frac{2+2}{\sqrt{4+12}+4} = \frac{4}{\sqrt{16}+4} = \frac{4}{4+4} = \frac{4}{8} = \frac{1}{2}$.

Answer: $\frac{1}{2}$.

$$17. \lim_{x \rightarrow 0} \frac{\sqrt{x+9}-3}{x}$$

Plugging in gives $\frac{0}{0}$. Rationalize with the conjugate $\sqrt{x+9}+3$:

$$\frac{\sqrt{x+9}-3}{x} \cdot \frac{\sqrt{x+9}+3}{\sqrt{x+9}+3} = \frac{(x+9)-9}{x(\sqrt{x+9}+3)} = \frac{x}{x(\sqrt{x+9}+3)}$$

Cancel x :

$$\frac{1}{\sqrt{x+9}+3}$$

Plug in $x=0$: $\frac{1}{\sqrt{9}+3} = \frac{1}{3+3} = \frac{1}{6}$.

Answer: $\frac{1}{6}$.

18. $\lim_{x \rightarrow 9} \frac{x-9}{\sqrt{x}-3}$

Plugging in gives $\frac{0}{0}$. Rationalize with the conjugate $\sqrt{x}+3$:

$$\frac{x-9}{\sqrt{x}-3} \cdot \frac{\sqrt{x}+3}{\sqrt{x}+3} = \frac{(x-9)(\sqrt{x}+3)}{x-9}$$

Cancel $(x-9)$:

$$\sqrt{x}+3$$

Plug in $x=9$: $\sqrt{9}+3 = 3+3 = 6$.

Answer: 6.

19. $\lim_{h \rightarrow 0} \frac{(2+h)^2-4}{h}$

Expand the top: $(2+h)^2 = 4+4h+h^2$, so the top becomes $4+4h+h^2-4 = 4h+h^2$.

$$\frac{4h+h^2}{h}$$

Cancel one h from each term:

$$4+h$$

Let $h \rightarrow 0$: $4+0 = 4$.

Answer: 4. (This is the exact same style of computation you used in Unit 1 to find a tangent slope — that's not a coincidence; it's the same skill in a new outfit.)

20. $\lim_{x \rightarrow 1} \frac{\frac{1}{x}-1}{x-1}$

The top is itself a small fraction problem. Combine $\frac{1}{x} - 1$ into a single fraction using a common denominator of x :

$$\frac{1}{x} - 1 = \frac{1-x}{x}$$

So the whole expression becomes:

$$\frac{\frac{1-x}{x}}{x-1} = \frac{1-x}{x(x-1)}$$

Notice that $1-x = -(x-1)$. Rewrite the top that way:

$$\frac{-(x-1)}{x(x-1)}$$

Cancel $(x-1)$:

$$\frac{-1}{x}$$

Plug in $x = 1$: $\frac{-1}{1} = -1$.

Answer: -1 .

Applied

21. $R(x) = 5x - 0.01x^2$, find $\lim_{x \rightarrow 10} R(x)$.

This is a polynomial, so direct substitution works — no tricks needed:

$$R(10) = 5(10) - 0.01(10)^2 = 50 - 0.01(100) = 50 - 1 = 49$$

Answer: 49. This means that as production gets closer and closer to 10 units, the revenue gets closer and closer to \$49.

22. $C(x) = 200 + 3x$. Find $\lim_{x \rightarrow 50} \frac{C(x)}{x}$.

At $x = 50$, the denominator is 50, which isn't zero, so direct substitution works:

$$\frac{C(50)}{50} = \frac{200 + 3(50)}{50} = \frac{200 + 150}{50} = \frac{350}{50} = 7$$

Answer: 7. As production nears 50 items, the average cost per item approaches \$7.

23. $P(t) = \frac{100t}{t+5}$. Find $\lim_{t \rightarrow 5} P(t)$.

At $t = 5$, the denominator is 10, which isn't zero, so direct substitution works:

$$\frac{100(5)}{5+5} = \frac{500}{10} = 50$$

Answer: 50 (thousand bacteria). As time nears 5 hours, the population approaches 50,000 bacteria.

24. $v(t) = \frac{t^2 - 4}{t - 2}$. Find $\lim_{t \rightarrow 2} v(t)$.

Plugging in $t = 2$ directly gives $\frac{0}{0}$, confirming the model formula itself breaks down at $t = 2$. But we can still find what it's approaching by factoring:

$$t^2 - 4 = (t - 2)(t + 2)$$

$$\frac{(t - 2)(t + 2)}{t - 2}$$

Cancel $(t - 2)$:

$$t + 2$$

Plug in $t = 2$: $2 + 2 = 4$.

Answer: 4. Even though the formula for $v(t)$ doesn't technically produce a value right at $t = 2$ (it's a $\frac{0}{0}$ gap in the model), the velocity is clearly heading toward 4 as t gets closer and closer to 2 from either side — so it's very reasonable to say the object's velocity is essentially 4 at that moment.